



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Create Short Video Film using AI

TGS Ref No: TGS-2020505925

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 12.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
10.0	Legacy release (superseded)	Legacy release delivered as 'Image and Video Processing with OpenCV' — 182 slides, six OpenCV topics, Raspberry Pi setup and Darknet/YOLO installation sequences. Retained as the content-coverage reference for this revision.	Dr. Alfred Ang
11.0	6 September 2026	Rebuilt as 'Generative AI for Image and Video Creation': five topics on the six approved Learning Units, twelve self-contained computer-vision and generative labs, all-white house design. Superseded by 12.0.	Dr. Alfred Ang
12.0	27 September 2026	Retitled 'Create Short Video Film using AI' and revamped for AI short-film production, following the five delivery topics now published on the course page. All six approved Learning Units, ELO1-ELO6, K1-K13, A1-A8 and the programme hours (450 CR + 360 PP + 150 AS minutes) are unchanged. New 138-slide deck and twelve new labs that build one 48-second film, 'The Last Ferry': need analysis, formats, structured prompting, state edits, grading, character consistency, look bible, cast detection, storyboard-to-shot prompts, continuity analytics, sound/captions/export and cloud-versus-local architecture. Workflow synthesised from seven current AI short-film tutorials (source register S6-S12). WA (13 questions, K1-K13, 60 min) and PP (5 tasks, A1-A8, 90 min) rewritten for the new scenario with counts, order, mapping and timings preserved.	Dr. Alfred Ang

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Course Information

Course Title	Create Short Video Film using AI
Registered Public Title	WSQ - Create Short Video Film using AI
WSQ Course Reference	TGS-2020505925
Skills Framework TSC	Computer Vision Technology (ICT-DIT-4022-1.1), Proficiency Level 4
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 16 programme hours (7.5 h classroom facilitation + 6 h practical + 2.5 h assessment)
Daily Timing	Day 1 9:00 am–6:30 pm; Day 2 9:00 am–6:40 pm. Administration from 8:45 am; 1-hour lunch and two 15-minute tea breaks are outside programme hours.
Mode	Instructor-led classroom facilitation with hands-on practical work; 12 self-contained labs building one short film, runnable locally with no paid service (AI generation tools optional)
Trainer	Dr. Alfred Ang
Trainer-to-learner ratio	1:3 to 1:10
Assessor-to-candidate ratio	1:3 to 1:10 for both WA(Q&A) and PP
Slide deck referenced	Create Short Video Film using AI-v12.0.pptx — 138 slides (version v12.0, 27 September 2026)

Programme-Hours Reconciliation and Time Assumptions

Planning timetable: preserve 450 minutes classroom facilitation, 360 minutes practical and 150 minutes assessment over two days, 480 programme minutes per day. Lunch, tea breaks and administration are additional, non-instructional time. Clock times below are a proposed trainer timetable, not a claim about a currently booked session. Each approved LU retains CR75 minutes and PP0/60/60/120/60/60 minutes respectively. Topic4 combines LU4 and LU5 but preserves both budgets. Lab01 is a facilitated needs-analysis discussion within LU1 classroom facilitation. Lab extensions are outside core class time. The public course page says two assessment hours; the authenticated LMS record and supplied proposal/assessment plan specify WA60 + PP90. The verified 150 minutes is used.

Component	Approved (min)	Scheduled (min)	Reconciles
Classroom facilitation	450	450	YES
Practical / practicum	360	360	YES
Assessment	150	150	YES
TOTAL PROGRAMME HOURS	960	960	YES

Note on the published assessment duration: the public course page states a 2-hour assessment. The registered Assessment Plan (TGS-2020505925 V1.0) and the approved Course Proposal (CA-WSQ-2020-013290-v2) both specify WA(Q&A) 60 minutes plus PP 90 minutes = 2.5 hours,

and that is what the authenticated LMS record carries. The 2.5-hour figure is authoritative and is used throughout this Lesson Plan.

Learning Outcomes and Competency Mapping

On completion of this course, learners will be able to:

- LO1: Understand basic vision systems concepts and applications
- LO2: Apply image processing
- LO3: Implement feature extraction
- LO4: Apply machine learning based computer vision methods
- LO5: Implement video analytics algorithms
- LO6: Evaluate edge vs cloud-based computer vision systems

ELO	Statement	Abilities	Knowledge	Delivery topic
ELO1	Understand basic vision systems concepts and applications	A1	K1, K2	Topic 01
ELO2	Apply image processing	A2	K3, K4	Topic 02
ELO3	Implement feature extraction	A3, A4	K5, K6, K7	Topic 03
ELO4	Apply machine learning based computer vision methods	A5	K8, K9	Topic 04
ELO5	Implement video analytics algorithms	A6	K10	Topic 04
ELO6	Evaluate edge vs cloud-based computer vision systems	A7, A8	K11, K12, K13	Topic 05

Assessment

- Written Assessment WA(Q&A) — 13 open-ended short-answer questions covering K1 to K13, 60 minutes, open book, assessor-to-candidate ratio 1:3 to 1:10.
- Practical Performance (PP) — 5 scenario-based tasks covering A1 to A8, 90 minutes, open book, assessor-to-candidate ratio 1:3 to 1:10.
- Total assessment time: 2.5 hours (150 minutes) on Day 2.
- Format: Open Book — course slides, Learner Guide and approved course materials only.
- Both instruments are conducted on Day 2 from 16:10, after the Briefing for Assessment and the Assessment digital attendance.
- An oral clarification of up to 5 minutes may be conducted for each instrument to close minor performance gaps; that time sits outside the assessment duration.
- A minimum of 75% attendance on the SSG digital attendance record is required, and candidates must be assessed as 'Competent' in every ability and knowledge statement to be eligible for WSQ funding.

Instrument	Item	Criterion	Practised in
WA(Q&A)	Question 1	K1	Lab 01
	Question 2	K2	Lab 01
	Question 3	K3	Lab 02, Lab 03
	Question 4	K4	Lab 03, Lab 04, Lab 05

Instrument	Item	Criterion	Practised in
	Question 5	K5	Lab 06
	Question 6	K6	Lab 06
	Question 7	K7	Lab 07
	Question 8	K8	Lab 08, Lab 09
	Question 9	K9	Lab 08, Lab 11
	Question 10	K10	Lab 09, Lab 10, Lab 11
	Question 11	K11	Lab 12
	Question 12	K12	Lab 12
	Question 13	K13	Lab 12
PP	Task 1	A1	Lab 01
	Task 2	A2, A3	Lab 02, Lab 03, Lab 04, Lab 05, Lab 07
	Task 3	A4	Lab 06
	Task 4	A5, A6	Lab 08, Lab 09, Lab 10, Lab 11
	Task 5	A7, A8	Lab 12

Course Schedule

Slide references below are generated from the built deck (Create Short Video Film using AI-v12.0.pptx, 138 slides) and are exact for this version. Any change to the deck regenerates this table.

Day 1 — Film concept, image generation and enhancement, and character and scene consistency

Time	Min	Component	Session	Slides
08:45–09:00	15	ADMIN	Course welcome, attendance, introductions and LMS access; outside programme minutes	—
09:00–10:15	75	CR	Topic1 / LU1: the AI film pipeline, how generators work, business uses; facilitated Lab01 film needs analysis (K1,K2; A1)	Slides 25–43
10:15–10:30	15	BREAK	Tea break; excluded	—
10:30–11:45	75	CR	Topic2 / LU2: frames and formats, structured prompting, state variants, reframing, grading and enhancement (K3,K4)	Slides 44–66
11:45–12:45	60	PP	LU2 practical: Labs02,03,04,05 core exercises, 15 minutes each; extensions completed after class (A2)	Slides 63, 64, 65, 66
12:45–13:45	60	LUNCH	Lunch; excluded	—
13:45–15:00	75	CR	Topic3 / LU3: character sheets, local and global features, look bible, shot design (K5,K6,K7)	Slides 67–85
15:00–15:15	15	BREAK	Tea break; excluded	—
15:15–16:15	60	PP	LU3 practical: Labs06 and07, 30 minutes each (A3,A4)	Slides 84, 85
16:15–17:30	75	CR	Topic4 / LU4: deep video models, shot prompting, continuity controls, cast detection and evaluation (K8,K9)	Slides 86–113
17:30–18:30	60	PP	LU4 practical part1: Lab08 cast-presence detection and evaluation (A5)	Slides 101

Day 1 programme time: 480 minutes (8 hours). Administration, tea breaks and lunch are additional and are shown for the trainer's timetable only.

Day 2 — Video generation and analytics, sound and finishing, cloud-versus-local, and assessment

Time	Min	Component	Session	Slides
08:45–09:00	15	ADMIN	Attendance and technical setup; outside programme minutes	—

Time	Min	Component	Session	Slides
09:00–10:00	60	PP	LU4 practical part2: Lab09 storyboard-to-shot prompts and Lab11 sound, captions and export, 30 minutes each (A5,A6)	Slides 102, 113
10:00–10:15	15	BREAK	Tea break; excluded	—
10:15–11:30	75	CR	Topic4 / LU5: temporal coherence, cuts, motion, tracking, drift and event analytics (K10)	Slides 86–113
11:30–12:30	60	PP	LU5 practical: Lab10 shot, motion and continuity analytics (A6)	Slides 112
12:30–13:30	60	LUNCH	Lunch; excluded	—
13:30–14:45	75	CR	Topic5 / LU6: production architectures, protocols, constraints and cloud-versus-local trade-offs (K11,K12,K13)	Slides 114–128
14:45–15:00	15	BREAK	Tea break; excluded	—
15:00–16:00	60	PP	LU6 practical: Lab12 cloud, local and hybrid production architecture evaluation (A7,A8)	Slides 128
16:00–16:10	10	ADMIN	Assessment briefing and attendance; outside programme minutes	—
16:10–17:10	60	AS	Written Assessment: 13 open-ended questions, K1-K13, individual/open book	Slides 135–137
17:10–18:40	90	AS	Practical Performance: five tasks, A1-A8, individual/open book; submission and sign-off	Slides 135–137

Day 2 programme time: 480 minutes (8 hours). Administration, tea breaks and lunch are additional and are shown for the trainer's timetable only.

Topic Detail and Slide Ranges

Topic	Learning Unit / ELO	K / A	Slides	CR	PP
01 AI Vision and Generative AI for Short Film Creation	LU1 · ELO1	K1, K2, A1	25–43	75	0
02 AI Image Processing, Generation and Visual Enhancement	LU2 · ELO2	K3, K4, A2	44–66	75	60
03 AI Visual Features, Character Consistency and Scene Design	LU3 · ELO3	K5, K6, K7, A3, A4	67–85	75	60
04 AI Video Generation, Animation and Video Analytics	LU4 + LU5 · ELO4 + ELO5	K8, K9, K10, A5, A6	86–113	150	180
05 Cloud and Edge AI Workflows for Short Film Production	LU6 · ELO6	K11, K12, K13, A7, A8	114–128	75	60

Hands-On Lab Reference

Every lab folder is independently usable: it carries its own numbered steps, mock data, local reference media, prompt pack, expected outputs, evidence checklist and troubleshooting table, and requires no paid service. Each learner-facing Markdown file has a same-basename PDF counterpart in the same folder.

Lab	Title	Topic	A	K	Min	Slide	Folder
01	Film Needs Analysis and AI Production Pipeline Specification	01	A1	K1, K2	in LU1	43	labs/lab-01-film-needs-analysis/
02	Frames, Formats and Aspect Ratios for Film Delivery	02	A2	K3	15	63	labs/lab-02-frames-formats-aspect-ratios/
03	Structured Prompting for Cinematic Stills	02	A2	K3, K4	15	64	labs/lab-03-structured-prompting-stills/
04	State Variants, Clean-Up and Reframing with Masks	02	A2	K4	15	65	labs/lab-04-state-variants-reframing/

Lab	Title	Topic	A	K	Min	Slide	Folder
05	Cinematic Grading and Enhancement Measured with PSNR and SSIM	02	A2	K4	15	66	labs/lab-05-grading-enhancement-metrics/
06	Character Reference Sheets and Consistency Features	03	A4	K5, K6	30	84	labs/lab-06-character-consistency-features/
07	Look Bible and Scene Continuity with Global Descriptors	03	A3	K7	30	85	labs/lab-07-look-bible-scene-continuity/
08	Cast-Presence Detection and Detector Evaluation	04	A5	K8, K9	60	101	labs/lab-08-cast-detection-evaluation/
09	Storyboard to Shot Prompts, Start and End Frames, and the Animatic	04	A5, A6	K8, K10	30	102	labs/lab-09-storyboard-shot-prompts/
10	Shot, Motion and Continuity Analytics on the Assembled Sequence	04	A6	K10	60	112	labs/lab-10-shot-motion-continuity-analytics/
11	Voiceover, Music, Captions and the Final Export Ladder	04	A5, A6	K9, K10	30	113	labs/lab-11-sound-captions-final-export/
12	Cloud, Local and Hybrid Production Architecture Evaluation	05	A7, A8	K11, K12, K13	60	128	labs/lab-12-cloud-local-production-architecture/

Instructional and Assessment Methods

Method	Ratio	Justification
Lecture	1:3 to 1:10	Transfers the underpinning knowledge for K1–K13 from facilitator to learners.
Didactic questioning	1:3 to 1:10	Confirms learners are following the explanation and can restate the mechanism, not just the term.
Demonstration	1:3 to 1:10	The facilitator runs each lab script once so learners see the expected output before attempting it.
Hands-on practical	1:3 to 1:10	Elicits performance for A1–A8 and lets the facilitator give feedback on the gaps each learner actually has.
WA(Q&A), 60 min	1:3 to 1:10	Direct written evidence that the learner holds the underpinning knowledge K1–K13.
Practical Performance, 90 min	1:3 to 1:10	Direct product evidence that the learner can perform A1–A8 on a scenario-based problem.

Resources Required

- One laptop per learner (Windows or macOS) with Python 3.10 or later, opencv-python 4.10 or later and NumPy installed before the class; FFmpeg optional (Lab 11 audio muxing).
- The labs/ folder, distributed from the LMS/TMS portal. All reference media is included locally; nothing is downloaded during the class.
- Projector and whiteboard.
- Slide deck: Create Short Video Film using AI-v12.0.pptx (138 slides) and its PDF.
- Learner Guide: LG-Create Short Video Film using AI.docx and its PDF and Markdown mirror.
- Assessment papers and answer keys (trainer copy only, not distributed).
- Internet access and AI tool accounts (image, video, voice, music) are OPTIONAL. They are used only for the optional generation extensions; every lab completes offline with the supplied SIMULATED media.

Source Register

Every external claim in the slides and the Learner Guide traces to one of the following. Vendor documentation was verified on 6 September 2026; model identifiers change, and the register records which were deprecated at that date.

Ref	Source	Locator
S1	Course Proposal Tertiary Infotech CA-WSQ-2020-013290-v2 (SSG-approved)	Part 4 Section E, Curriculum Key Features table
S2	Assessment Plan_OpenCV_TGS-2020505925_V1.0 (28 Dec 2020)	Assessment Duration table; WA and PP specification tables
S3	Competency Mapping.docx (TSC ICT-DIT-4022-1.1)	Course Mapping section and ELO/A-code matrix
S4	Computer Vision Technology.docx (Skills Framework TSC extract)	TSC Proficiency Level 4 (ICT-DIT-4022-1.1) knowledge and abilities columns
S5	Tertiary Courses public registration page and LMS-TMS course record	https://www.tertiarycourses.com.sg/wsqa-create-short-video-film-using-ai.html (checked 27 Sep 2026)
S6	Youri van Hofwegen, 'EXACTLY How to Make an AI Short Film (Full Workflow)', YouTube, 1 Jun 2026, youtube.com/watch?v=0v534yAyhgw	Full video, 16:49
S7	Mira AI, '6 AI Video Tools I'd Use To Make An AI Short Film', YouTube, 14 Jun 2026, youtube.com/watch?v=Whh8vCqboxQ	Full video, 10:04
S8	Thomas Creates, 'How to Start Making AI Short Films in 2026', YouTube, 18 Sep 2026, youtube.com/watch?v=69_EJD4FEPg	Video description and chapters, 12:42
S9	Roboverse, 'The Complete Guide to Making Cinematic AI Videos (2026)', YouTube, 19 Jul 2026, youtube.com/watch?v=HMVSCEh72n4	Chapters, 10:30
S10	Youri van Hofwegen, 'How to Make Your First AI Movie (Full Guide)', YouTube, 11 Jun 2026, youtube.com/watch?v=fs5S867VQzg	Video description, 17:03
S11	ElevenLabs, 'How to Make Your First AI Short Film (Full Tutorial)', YouTube, 9 Aug 2026, youtube.com/watch?v=47DH_VUa67A	Chapters, 31:13
S12	Sanky, 'How I Made an AI Short Drama in under 10 Minutes with ZooClaw + Seedance2', YouTube, 2 Jun 2026, youtube.com/watch?v=q8z5iK5Pm_s	Chapters and description, 7:47
S13	M. Tschochohei and F. Schenker, 'Multimodal Generative AI in the Enterprise: From Pixels to Profit', Packt, June 2026, ISBN 978-1-80611-167-1	Ch.5 pp.56-75; Ch.7 pp.106-119; Ch.8 pp.127-143
S14	A. Mewada, M. A. Ansari, S. Ahmad and N. Singh (eds), 'AI-Generated Image and Video Synthesis', IEEE Press / Wiley, 2026, ISBN 9781394403110	Ch.1 pp.1-11; Ch.3 pp.35-44; Ch.7 pp.94-100
S15	Google AI for Developers — Gemini API image generation documentation	https://ai.google.dev/gemini-api/docs/image-generation (verified 6 Sep 2026)
S16	Google AI for Developers — Veo video generation documentation	https://ai.google.dev/gemini-api/docs/veo (verified 6 Sep 2026)
S17	OpenCV 4.13 Python package and FFmpeg, verified locally on the delivery image	cv2 4.13.0 (opencv-python 4.13.0.92); ffmpeg/ffprobe optional